# Sardar Vallabhbhai National Institute of Technology, Surat

**Mathematics Department**

**B.Tech-1 Subject: Mathematics I**

**Tutorial - 06 : Series solution of ODE with regular singular point**

**Find the series solution of the following differential equations:**

**1.** $4x\frac{d^2y}{dx^2}+2\frac{dy}{dx}+y=0$

**Ans**: $y=A\cos\sqrt{x}+B\sin\sqrt{x}$ , *where* $A=c_1a_0$ , $B=c_2a_0$

**2.** $2x(1-x)y''+(1-x)y'+3y=0$ *about* $x_0=0$

**Ans**: $y=A\left(1-3x+x^2-...\right)+B\sqrt{x}\,(1-x)$, *where* $A=c_1a_0$ , $B=c_2a_0$

**3.** $xy''+y'-y=0$ *about* $x_0=0$

**Ans**: $y=(A+B\log x)\left(1+x+\frac{x^2}{4}+...\right)-2B\left(x+\frac{3}{8}x^2+...\right)$, *where* $A=c_1a_0$ , $B=c_2a_0$

**4.** $x(x-1)y''+(3x-1)y'+y=0$ *at* $x=0$

**Ans**: $y=(A+B\log x)\,(1-x)^{-1}$ , *where* $A=c_1a_0$ , $B=c_2a_0$

**5.** $xy''+y'+xy=0$

**Ans**: $y=(A+B\log x)\left(1-\frac{x^2}{2^2}+\frac{x^4}{2^2\cdot4^2}-...\right)+B\left(1+\frac{x^2}{2^2}-\frac{3}{2^3\cdot4^2}x^4+...\right)$, *where* $A=c_1a_0$ , $B=c_2a_0$

**6.** $x^2y''+x^3y'+(x^2-2)y=0$

**Ans**: $y=\frac{A}{x}+Bx^2\left(1-\frac{3}{10}x^2+\frac{3}{56}x^4-...\right)$, *where* $A=c_1a_0$ , $B=c_2a_0$

**7.** $(x^2-x)y''-xy'+y=0$ *about* $x=0$

**Ans**: $y=(A+B\log x)+B(1-3x)$, *where* $A=c_1c_0$ , $B=c_2c_0$

**8.** $x^2y''+xy'+(x^2-n^2)y=0$

**Ans**: $y=a_0x^m\left(1-\frac{1}{(m+2)^2-n^2}x^2+\frac{1}{\left[(m+2)^2-n^2\right]\left[(m+4)^2-n^2\right]}x^4-...\right)$

**9.** $x^2y''+xy'+(x^2-1)y=0$

**Ans**: $y=(A+B\log x)\left(x-\frac{x^3}{2\cdot4}+...\right)+\frac{B}{x}\left[1+\frac{x^2}{4}-\left(\frac{1}{64}+\frac{2}{32}\right)x^4+....\right]$

**10.** $(1-x^2)y''-2xy'+n(n+1)y=0$

$y(x)=a_0y_1(x)+a_1y_2(x)$ *where* $y_1(x)=1-\frac{n(n+1)}{2}x^2+\frac{(n-2)n(n+1)(n+3)}{24}x^4-...$ *and*

**Ans**:

$y_2(x)=x-\frac{(n-1)(n+2)}{6}x^3+\frac{(n-3)(n-1)(n+2)(n+4)}{120}x^5-...$